

Reuben Rajan George

Principal Architect — System Performance & Reliability

2K Cross, 9th Main, HRBR 1st Layout, Bengaluru, Karnataka 560043

+91-9036302719 | reubenrajan@gmail.com

linkedin.com/in/reubenrajan | reubenrajan.dev | research.reubenrajan.dev | projects.reubenrajan.dev

INTRODUCTION

Principal Architect with 15 years of hands-on experience making complex systems faster, more resilient, and more observable, across distributed cloud architectures, blockchain platforms, IoT infrastructure, SAP, and digital and mobile ecosystems. The experience and lessons learnt is now directly applied to the most challenging reliability problems in enterprise AI: evaluating agentic systems, optimising inference cost and latency, and building the frameworks organisations need to make agentic AI efficient, trustworthy, cost-efficient, and production-ready.

SKILLS

- **Develop:** JavaScript, Python, AI Assistants (Claude, Github Copilot, Antigravity), LangGraph
- **Build & Deploy:** Gitlab, JIRA, Confluence
- **Test — Performance Engineering:** K6, JMeter, Artillery, LoadRunner, NeoLoad
- **Test — Chaos & Resilience:** Harness, Gremlin, ChaosToolkit, LitmusChaos
- **Test — AI Evaluation:** DeepEval, MLflow
- **Observe — Systems:** Dynatrace, Splunk, Grafana, Prometheus, OpenTelemetry, NewRelic, DataDog
- **Observe — AI Systems:** MLflow, DeepEval, OpenTelemetry
- **Infrastructure:** AWS, Kubernetes, Fargate, Vercel, CloudFlare Workers

CERTIFICATIONS

- Claude Certified Architect - Associate (Pursuing)
- AWS Solution Architect Associate
- Gremlin Certified Chaos Engineering Professional
- Harness Certified Expert - Resilience Testing
- Reinvention with Agentic AI (Accenture Forward Deployed Engineer Course)
- Neoload Certified Expert

EXPERIENCE

Accenture Solutions Pvt. Ltd — AI & Agentic Quality January 2026 – Present | Bengaluru

- Developing Accenture's agentic evaluation and well-architected frameworks to benchmark, assess, and optimize performance, reliability, and cost of agentic AI systems at enterprise scale.
- Built Claude-powered agentic test automation agents for end-to-end performance testing of packaged enterprise applications (SAP, Oracle ERP) — reducing manual test effort and enabling continuous reliability validation.
- Part of Accenture's AI Quality Advisory Team, shaping guidance on AI system quality, safety, evaluation standards, and responsible agentic deployment practices.

Accenture Solutions Pvt. Ltd — Principal Architect, System Reliability Engineering December 2018 – December 2025 | Bengaluru

- Built Accenture's Reliability Testing practice from the ground up — defined standards, tool frameworks, and delivery frameworks adopted across multiple enterprise engagements.

- Architected and deployed self-managed Chaos Engineering platforms for enterprise clients using Gremlin, ChaosToolkit, and LitmusChaos on AWS and Kubernetes environments.
- Designed full-stack observability solutions using Thanos, Prometheus, Grafana, and OTel; built custom dashboards and alert frameworks, with notification webhooks.
- Led performance engineering engagements using JMeter, Gatling, LoadRunner, and NeoLoad — covering workload modelling, bottleneck analysis, and capacity planning across distributed cloud architectures.
- Built a custom performance testing framework to benchmark and tune a Hyperledger Fabric-based CBDC platform for a national central bank — tuned thread pooling, JVM configuration and optimised transaction throughput for high-frequency wallet token transactions.
- Traced end-to-end transaction journeys across application, infrastructure, and network layers to pinpoint bottlenecks using AppDynamics, NewRelic, and Dynatrace.
- Helped establish SRE as an internal discipline at Accenture — developed course curriculum, ran workshops, and contributed to thought leadership through conference talks and white papers.

Accenture Solutions Pvt. Ltd — Digital Performance Test Architect August 2015 – November 2018 | Bengaluru

- Founded and led Accenture's Digital Performance Engineering capability covering mobile (Android, iOS) and browser performance, defining testing methodologies, toolchains, and optimization playbooks from scratch.
- Built frameworks to profile and optimize mobile app performance using Android Profiler, Xcode Instruments, and Charles Proxy; analyzed rendering, memory, CPU, and network behaviour to identify and resolve performance regressions.
- Developed web benchmark frameworks using WebPageTest, Lighthouse, and Chrome DevTools — measuring and improving Core Web Vitals, Time to Interactive, and render-critical resource loading; deployed at scale for a Google Research Study benchmarking 600+ retail, travel and banking websites across the APAC region.
- Served as SME on mobile and digital performance proposals for 150+ enterprise clients, shaping testing strategy, tool selection, and SLA frameworks across diverse technology stacks.
- Delivered 8 digital performance engineering engagements spanning mobile app optimization and system-level performance validation; accountable end-to-end from scoping through root cause analysis.
- Designed and executed performance tests for IoT messaging frameworks using MQTT and AMQP protocols; simulated high-concurrency device telemetry workloads using JMeter MQTT plugin, validated message throughput and broker saturation thresholds under peak load.
- Established technology partnerships to expand Accenture's performance testing offerings and co-develop client-facing GTM solutions.
- Built and scaled Performance Test & Automation Labs across 7 Accenture delivery locations in India and globally (Mexico, Brazil, Italy, Philippines, China), standardizing tooling, environments, and training curricula.

Accenture Solutions Pvt. Ltd — Performance Testing Lead, Global Hospitality Company July 2014 – August 2015 | Bengaluru

- Led a distributed, multi-location SAP performance test team to validate infrastructure refresh across 14+ SAP systems including ECC, BW, and CRM.
- Scripted and executed SAP performance tests using HP LoadRunner with SAP-Web and SAP-GUI protocols; monitored system behaviour through SAP transaction codes (ST03, STAD) to analyse work process utilisation, lock contention, and dialog step response times.
- Monitored end-to-end transaction performance using Wily Introscope (CA APM).
- Established pre-migration performance baselines and ran post-migration benchmarks to validate SLA compliance and quantify infrastructure gain — providing sign-off evidence for production cutover.
- Received Accenture Excellence Award for on-time delivery and client-cited performance improvements following infrastructure refresh.

Accenture Solutions Pvt. Ltd — Performance Test Engineer, Global Pharmaceutical Company June 2011 – July 2014 | Bengaluru

- Delivered performance testing across 6 digital marketing projects for a global life sciences company — covering drug brand marketing websites, prescription product landing pages, HCP portals, patient support platforms, and medical information web applications.
- Performance tested SOA-based integration layers using HP LoadRunner with Web Services (SOAP/REST) protocols; validated message throughput, service response SLAs, and transaction throughput.
- Supported test automation activities using HP Quality Center (ALM) for defect tracking, test planning, and compliance-aligned test evidence management across multiple concurrent project tracks.

THOUGHT LEADERSHIP

May 2025 — MIT STAMP Workshop, Boston — "Application of STAMP/STPA in Distributed Digital Infrastructure – eCommerce System Example": Applied MIT's System-Theoretic Accident Model and STPA hazard analysis methodology to identify systemic safety risks and emergent failure modes in large-scale distributed eCommerce architectures.

January 2025 — Chaos Carnival 2025 (Harness) — Speaker: "Resilience and Chaos Engineering Automation"

May 2022 — ACM/IEEE ICSE 2022 — Industry whitepaper "Improved Business Outcomes from Cloud Applications – using Integrated Process and Runtime Product Data Mining": Demonstrated how combining process telemetry with runtime product data yields actionable insights for improving cloud application outcomes.

2021 — Software Testing Conference (STC 2021) — Best of the Best Award for industry paper "Continuous System Resilience Management": Recognised for advancing the discipline of automated resilience validation integrated into the software delivery lifecycle.

March 2021 — Conf42 Chaos Engineering — Lightning Talk: "Role of Quality Engineers in SRE": Addressed quality engineers transitioning into SRE, covering observability within testing pipelines and automating chaos scenario creation for pre-production environments.

February 2021 — Singapore Government — Workshop: "UX Master Class Series on SEO & Performance": Covered web performance engineering including Core Web Vitals optimisation, SEO-performance intersection, and measurement frameworks for government digital services.

June 2020 — IEEE/ACM ICGSE 2020, Seoul — Industry whitepaper "Observability Driven Development and Continuous Delivery": Proposed embedding observability instrumentation as a first-class practice within CI/CD pipelines rather than a post-deployment concern.

September 2019 — Performance Advisory Council, Bangalore — Talk: "Art of Observability": Covered observability strategies for distributed systems including structured logging, distributed tracing, and metrics-driven performance engineering.

2018 — Google "Masters of Mobile" APAC Research (Think with Google) — Built the performance benchmarking framework and designed the evaluation methodology; the study assessed 700+ mobile sites across retail, travel, and financial services brands across APAC, Australia, and New Zealand on mobile UX, page speed, and conversion-impacting performance metrics.

PATENTS

US10459835B1 — System and Method for Controlling Quality of Performance of Digital Applications (Granted October 2019)

An automated performance engineering system that uses AI-trained bots to test web and mobile apps under simulated real-world conditions (network, CPU, memory, battery). Detects performance issues, classifies them by

urgency, auto-applies front-end remediations, and validates improvements via before/after metric comparison before production deployment.

US20230315954A1 — Dynamic Failure Mode Effect Analysis and Recovery Process Recommendation for Cloud Computing Applications (Granted October 2023)

A dynamic risk analysis engine for cloud-native and IoT/edge systems that uses ML to identify failure patterns from historical operational metadata. Assigns Risk Priority Numbers (RPN) based on severity, probability, and detectability, then recommends context-aware recovery processes optimised for minimal latency and compute overhead.

US20240168857A1 — Utilizing Digital Twins for Data-Driven Risk Identification and Root Cause Analysis of a Distributed and Heterogeneous System (Granted May 19, 2026)

A digital twin architecture for hybrid cloud environments where individual component twins feed operational metrics into a central coordinating twin. Uses PCA (reconstruction error analysis) and Self-Organizing Maps (SOM) for anomaly detection, then generates a KPI cause vector to trace root causes iteratively across causality levels.

US20240403135A1 — Cloud Fitness Engineering (Published December 2024, Pending)

An ML-driven system that identifies optimal cloud architectures for specific applications by decomposing business performance goals into measurable technical parameters, then simulating candidate architectures under stress conditions. A fitness design controller selects the architecture that best satisfies competing constraints and trade-offs.

AWARDS

- Accenture Celebrates Excellence Awards for exceptional contribution in Innovation & Project Delivery — 2025, 2022, 2021, 2020, 2019, 2018, 2016
- APAC Region Award for delivery project excellence in driving change in business innovation — 2015 (Google Research)
- Life Sciences Delivery Excellence Award for exceptional project contribution — 2014

EDUCATION

Life Christian University — Bachelor of Theology August 2021 – August 2025 | Florida, USA

Mahatma Gandhi University — Bachelor of Technology, Information Technology August 2007 – June 2011 | Kerala, India

- Passed with 71% GPA
- Project: Developed a Human-Computer Interface where the computer mouse was controlled by an eye-tracking algorithm built on Matlab.
- Research Paper: Applying data mining techniques to determine crime patterns facilitating law enforcement to identify crime sprees, crime types, and organized criminal gangs.
- Point Guard for college Basketball team, competed in inter-college tournaments.

Indian Community School Kuwait — CBSE 10th & 12th Board March 2005 – March 2007 | Kuwait

- Completed both 10th & 12th Boards with distinction.
- Awarded for being among top rank holders in Kuwait for overall score & mathematics.
- School Prefect.

The above information provided is correct to the best of my knowledge and belief.